

The Power of Vaccination: From Smallpox Eradication to COVID-19 in

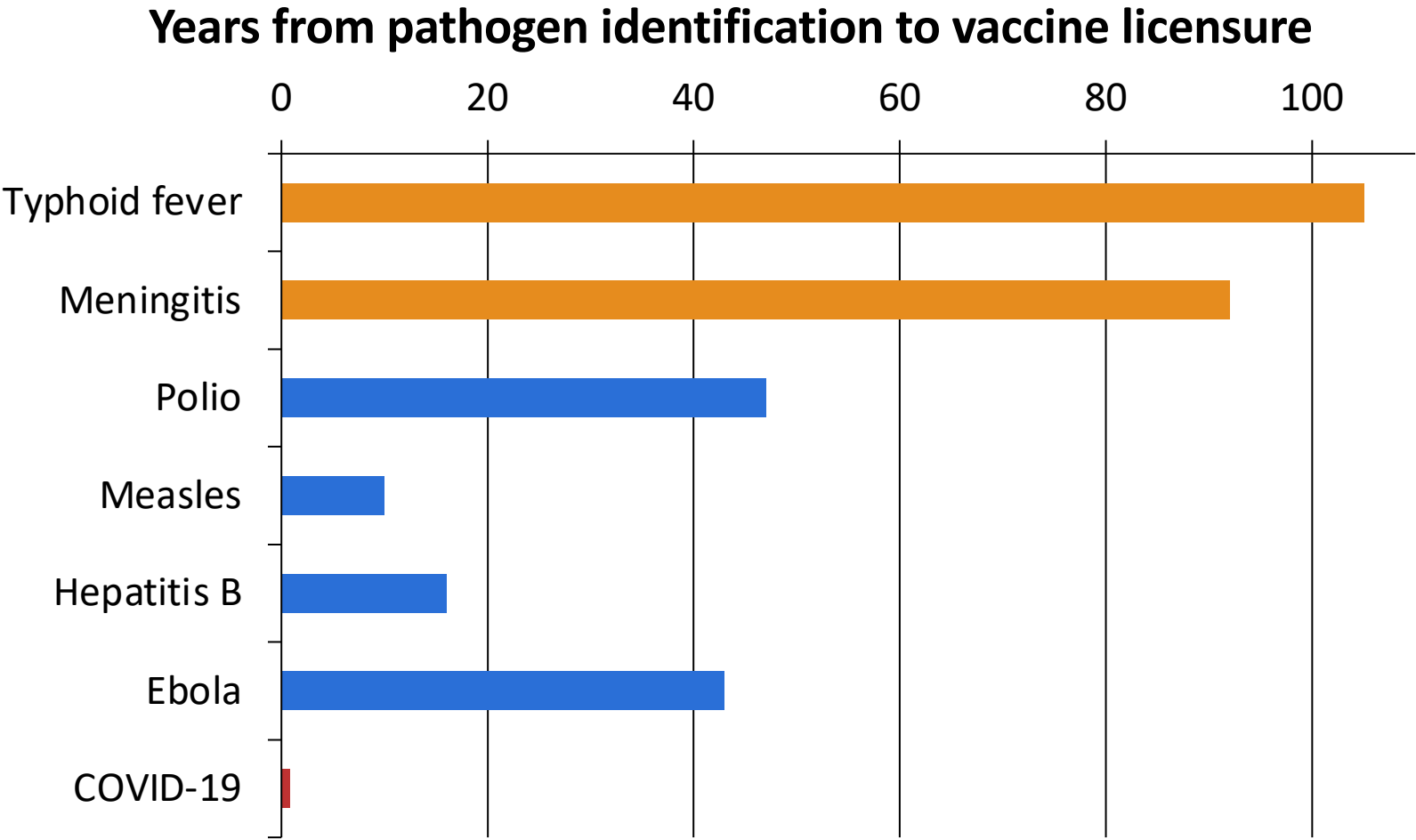
A data-driven overview of vaccine progress, impact, and remaining gaps (1796–2020)

Audience: Public health professionals and policy analysts

Source: Our World in Data (vaccination), with disease reduction data from Roush & Murphy (2007)

Scope: historical vaccine development timelines, US disease elimination data, measles state trends, herd immunity, and remaining challenges.

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



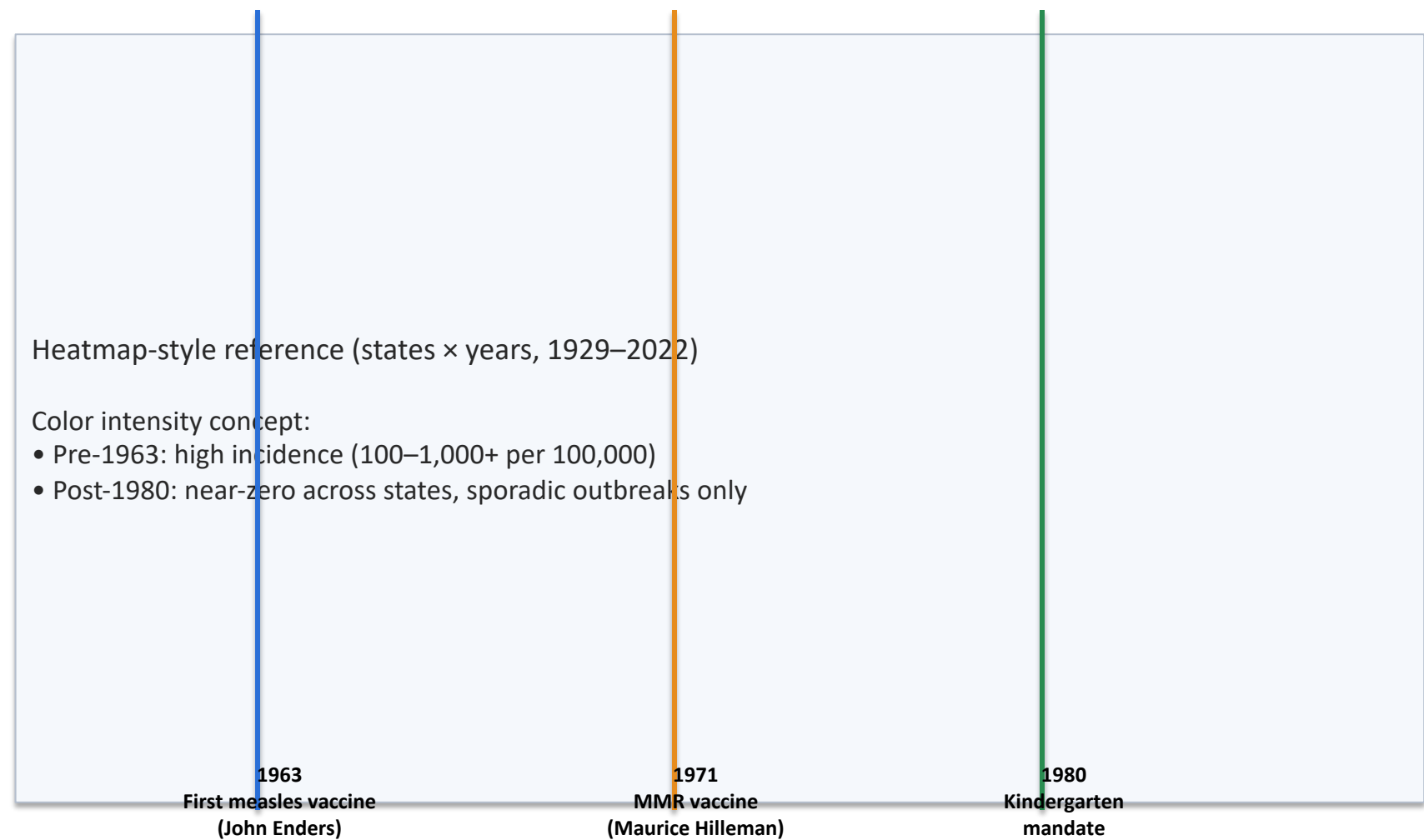
Key signal

- Typhoid: 105 years (1884 → 1989)
- Meningitis: 92 years (1889 → 1981)
- Measles: 10 years (1953 → 1963)
- COVID-19: <1 year (2020 → 2020)
- Acceleration enabled by genome sequencing, mRNA platforms, and advanced microscopy.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in

Disease	Pre-vaccine cases (per million/yr)	Post-vaccine cases (per million/yr)	Case reduction	Pre-vaccine deaths (per million/yr)	Death reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

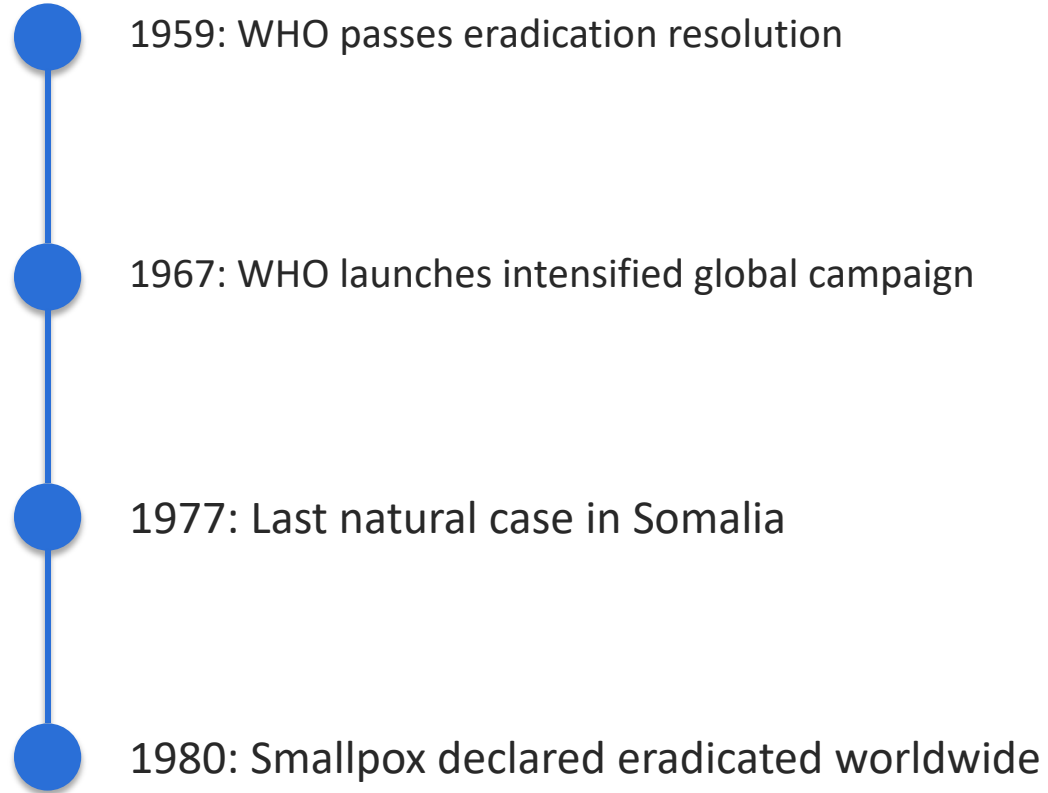
Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Interpretation

- Before vaccination, nearly every state had su
- After 1963 and especially after the 1980 scho
- Pattern demonstrates durable policy + vaccin

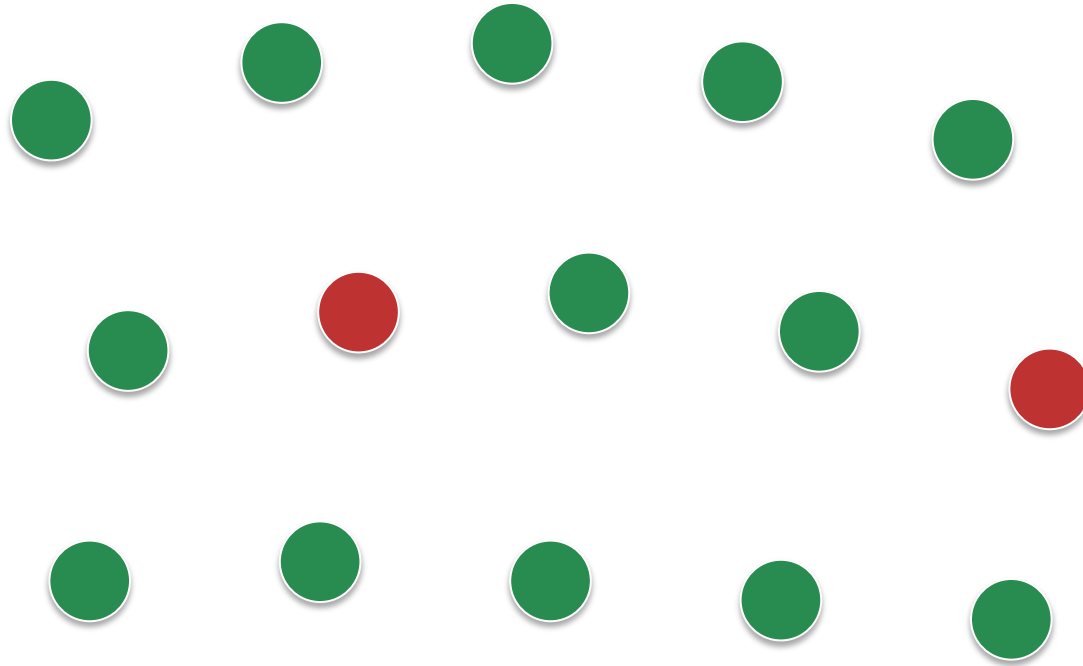
Smallpox: The Only Human Disease Eradicated Through Vaccination



Why eradication was possible

- Virus infected only humans
- Symptoms were clearly distinguishable
- Vaccine was cheap and effective
- Global coordination sustained over decades

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



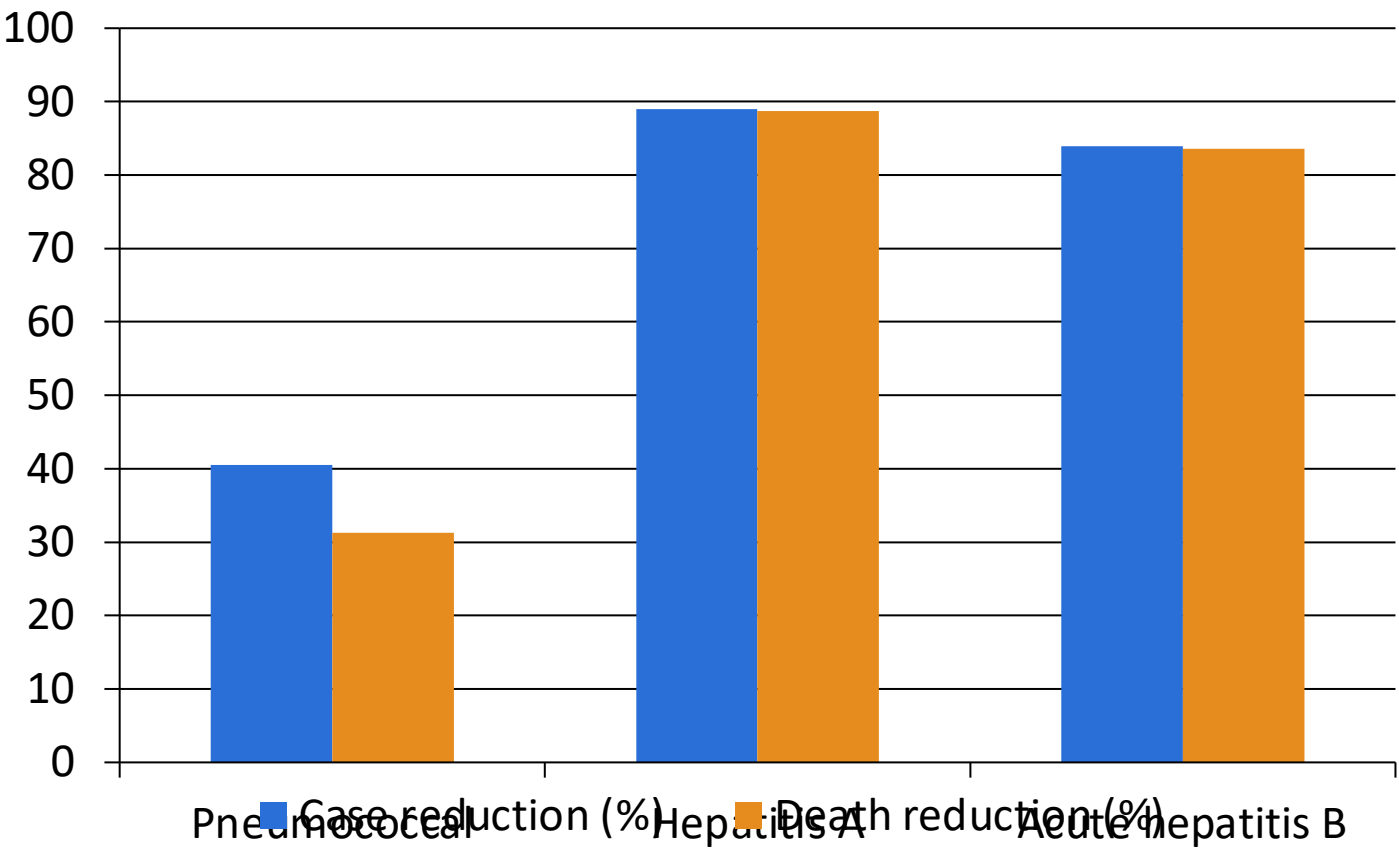
● Green: vaccinated/protected ● Red: unvaccinated

When most nodes are vaccinated, transmission chains break and vulnerable individuals are indirectly

Who benefits most from indirect protection?

- Newborns too young for some vaccines
- Immunocompromised individuals
- Cancer patients on chemotherapy

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Persistent barriers

- Incomplete control for selected diseases despite vaccination
- Global access gaps leave many populations under-immunized
- Supply shortages put children at risk
- Weak scientific understanding can reduce trust and uptake

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the

What this means for health systems:

- Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and acute polio in the US.
- Development timelines collapsed from over 100 years (typhoid: 105 years) to under 1 year (COVID-19).
- Measles incidence fell from widespread high state-level rates to near zero after vaccine and policy milestones.
- Smallpox remains the only human disease eradicated through vaccination (certified in 1980).
- Coverage gaps persist for pneumococcal disease, hepatitis A, and acute hepatitis B.

Call to action: strengthen vaccine access, coverage, and trust with sustained policy, financing, and delivery infrastructure.